

12.159

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Question:

The determinant

$$\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (0.1)$$

evaluates to

- a) 0
- b) $2b(b-1)$
- c) $2(1-b)(1+2b)$
- d) $3b(1+b)$

Solution:

Let the determinant of the given matrix be Δ .

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (4.1)$$

We will use row operations to simplify the determinant. Applying the operation $R_2 \rightarrow R_2 - R_1$ does not change the value of the determinant.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b-(1+b) & 1+b-b & 1-1 \\ 1 & 2b & 1 \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 1 & 2b & 1 \end{vmatrix} \quad (4.2)$$

Next, applying the operation $R_3 \rightarrow R_3 - R_1$ also does not change the value of the determinant.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 1-(1+b) & 2b-b & 1-1 \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ -b & b & 0 \end{vmatrix} \quad (4.3)$$

Now that the third column is in the form $(1, 0, 0)^T$, we can easily evaluate the determinant by expanding along this column:

$$\Delta = 1 \cdot \begin{vmatrix} -1 & 1 \\ -b & b \end{vmatrix} - 0 + 0 \quad (4.4)$$

$$= ((-1)(b) - (1)(-b)) \quad (4.5)$$

$$= -b + b = 0 \quad (4.6)$$

The determinant is 0, which corresponds to option (a).